

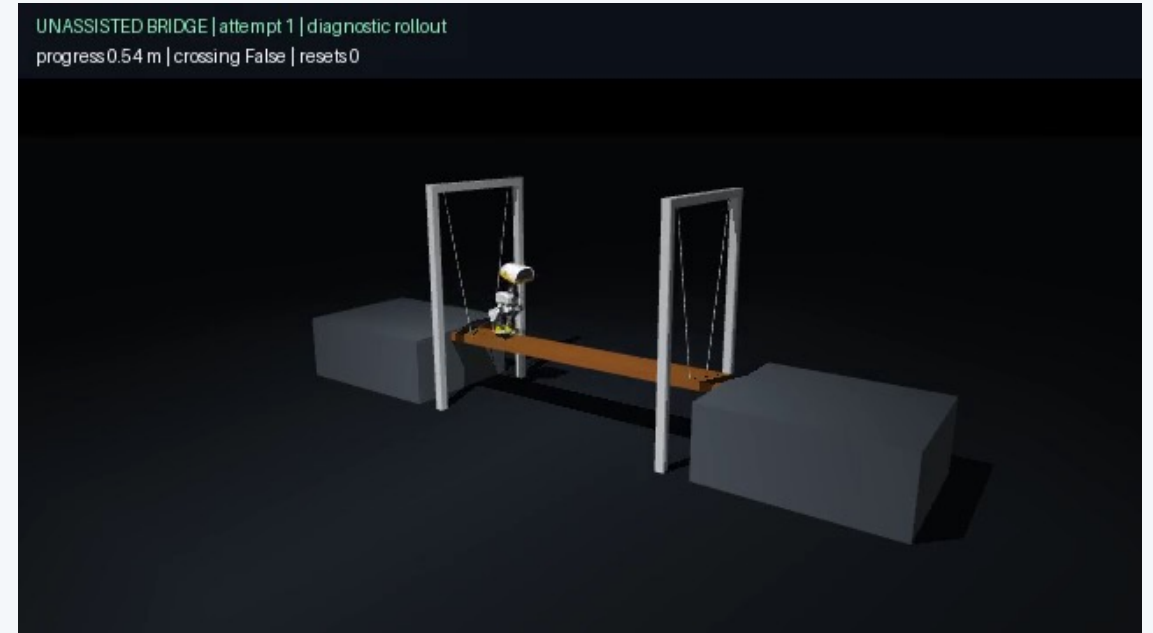
LAB 07

Suspended bridge

Suspended Bridge: A Failed Crossing Worth Studying

Environment design · PPO · Honest verification

BUILD → TRAIN → VERIFY → EXPLAIN



What you will build — and what the evidence says

- 01** Use case: Teach moving-support simulation, privileged command generation, physics-only curricula, warm-start PPO, strict contact-based evaluation, and scientific reporting of an unsuccessful policy. The environment and pipeline work, but the retained pilot did not complete an accepted crossing.
- 02** By the end: describe the scene, map observations to actions, explain the reward, trace one PPO update, run a bounded experiment, and defend an evaluation claim.
- 03** The bridge crossing failed. Partial progress and a valid scene do not establish the requested skill.
- 04** Scope: local MuJoCo simulation. Do not deploy these classroom artifacts to a physical robot; hardware transfer requires a separate validated process.

Materials and prerequisites · 1/3

- 01 Microduck MuJoCo model with XML position servos and the standard alpha_walking.onnx bootstrap
- 02 One free plank, 1.10 x 0.13 x 0.03 m, mass 0.45 kg
- 03 Two fixed platforms, each 0.60 x 0.48 x 0.24 m
- 04 Four spatial tendon cables with stiffness 320, damping 2.0, width 0.0015 m, and fixed anchors

Materials and prerequisites · 2/3

- 05 Visible non-colliding gantry posts and crossbars plus a separate ground plane for failure detection
- 06 `rlx/rlx/environments/bridge.py` for geometry, commander, assistance, contacts, metrics, and curriculum
- 07 `rlx/rlx/environments/bridge_evaluation.py` for the full-horizon contact-validated crossing gate
- 08 `rlx/rlx/models/bridge_bootstrap.py` for exact walking actor and normalizer transfer

Materials and prerequisites · 3/3

09

`rlx/examples/ppo_microduck_studio.py` and `scripts/train-bridge.sh` for training, evaluation, rendering, and export

10

`rlx/runs/studio/bridge/bridge-studio-02/` for the retained 32,768-transition pilot and failed diagnostic video

Environment and policy contract · 1/3

- 01** The plank is a MuJoCo free body centered initially at (0,0,0.225) m. Four physical tendons connect plank attachment sites to anchors at $x \pm 0.45$ m, $y \pm 0.17$ m, $z = 0.77$ m.
- 02** The nominal plank spans $x = -0.55$ to 0.55 m. Start and end platforms are centered at $x = -0.85$ m and $x = 0.85$ m, with all top surfaces at $z = 0.24$ m.
- 03** The visible gantry uses $\text{contype} = 0$ and $\text{conaffinity} = 0$. It cannot support or rescue the robot; only tendons, platforms, plank, and ground participate in contact dynamics.
- 04** Physics runs at 0.005 s with four substeps per 14-action command, giving a 0.02 s control interval and 50 Hz.

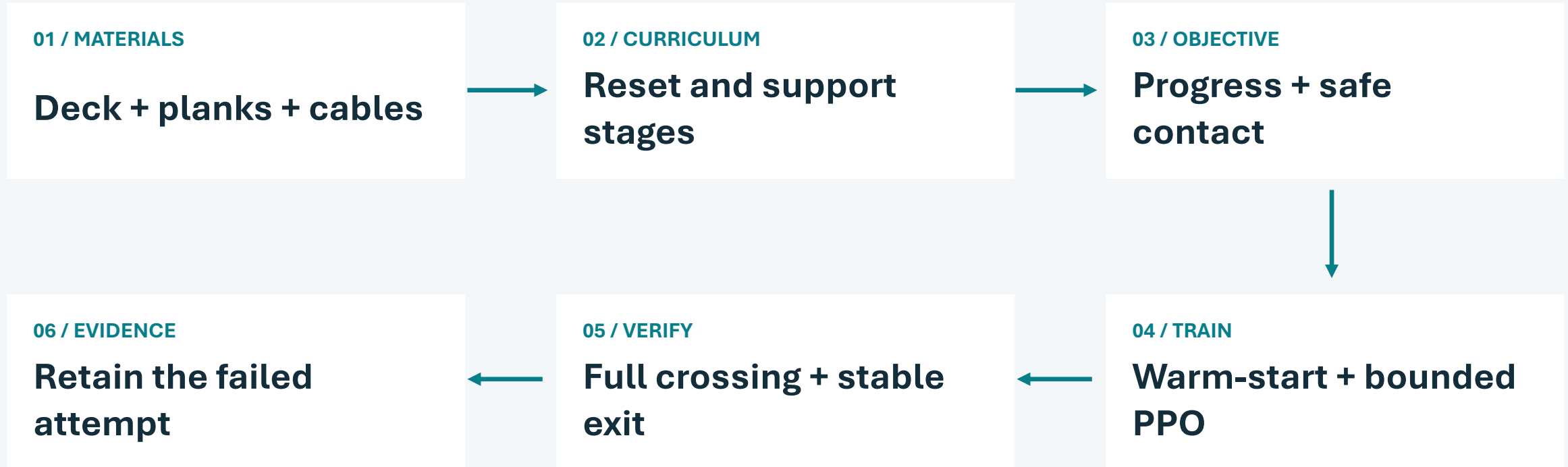
Environment and policy contract · 2/3

- 05** Observation/action contract remains float32[61] to 14 joint-target offsets. Observation slices are angular velocity 0:3, projected gravity 3:6, joint offsets 6:20, lagged joint velocity 20:34, previous action 34:48, twist command 48:51, zero head command 51:55, and zero body command 55:61.
- 06** The actor receives no plank position, velocity, roll, pitch, cable state, world position, bridge contacts, crossing flag, or assistance level.
- 07** The external navigation commander uses world trunk x/y and heading to publish the ordinary body-frame twist command. It commands nominal 0.25 m/s forward, lateral correction clip(-1.2*y, -0.12, 0.12), and yaw correction clip(-1.5*yaw_error, -0.6, 0.6).
- 08** At world x at least 0.80 m, the commander sets desired forward velocity to zero while retaining lateral and yaw correction.

Environment and policy contract · 3/3

- 09 Physics-only assistance levels are 1.0, 0.6, 0.3, 0.1, and 0.0. They scale restoring force, leveling torque, and damping on the plank, never on the robot.
- 10 Curriculum rungs also change declared reset spawn distance. Reward weights remain identical. Evaluation and rendering use assistance 0 on the final launch-platform spawn.
- 11 An accepted crossing requires at least 1.60 m progress from launch, prior foot contact with the suspended plank, both feet on the end platform, no feet or body on the ground, no non-foot support, no assistance, no fall, and the complete 1,000-step horizon.

How this case is built: one evidence loop



Failure is data: change one cause, retrain, and repeat the same evaluation gates.

Build the case step by step · 1/3

- 01** Construct the free plank, platforms, four tendons, anchors, non-colliding gantry, and failure-detection floor from the recorded dimensions.
- 02** Verify that moving the plank without moving the robot does not change the 61-value actor observation.
- 03** Load `alpha_walking.onnx`, transfer its actor and normalizer exactly, check action parity, freeze observation normalization, and create a fresh critic and optimizer.
- 04** Enable bridge-curriculum only for training so early rollouts can sample stabilized planks and closer spawns.

Build the case step by step · 2/3

- 05 Keep the inherited locomotion reward unchanged; do not add world-x progress or bridge-crossed jackpots.
- 06 Train 4 environments in complete 128-step batches until 32,768 new transitions are collected.
- 07 Export bridge.onnx with the frozen normalizer baked into the standard raw 61-input interface.
- 08 Evaluate 3 lanes for 1,000 steps each with seed 101, no assistance, no randomization, no observation noise, no action delay, and no random yaw.

Build the case step by step · 3/3

09

Track each reset-produced attempt and trailing partial fragment separately; do not reinterpret 3,000 lane transitions as three completed successful episodes.

10

Apply the contact-validated crossing gate and retain failed episodes, floor contacts, missing plank contacts, progress, and support counts.

11

Render the complete 20-second diagnostic and mark its two resets so partial attempts are not presented as one continuous crossing.

PPO: improve the actor without trusting a huge step

$$\text{ratio} = \pi_{\text{new}}(a \mid o) / \pi_{\text{old}}(a \mid o)$$

Maximize $\text{mean}[\min(\text{ratio} \times A, \text{clip}(\text{ratio}, 1 - \epsilon, 1 + \epsilon) \times A)]$

Actor: chooses joint actions. Critic: estimates future discounted reward.

A = advantage. ϵ = clipping range. Old log probabilities stay fixed for each collected batch.

Follow one rollout through the learner

- 01** Collect observations, sampled actions, old log probabilities, values, rewards and episode-boundary flags using the current stochastic actor.
- 02** Compute TD residuals and generalized advantage estimates (GAE). A true terminal has zero continuation value; a timeout may bootstrap the saved final observation.
- 03** Cut advantage traces at reset boundaries. Never bootstrap a timeout from the first observation of the next episode.
- 04** Reuse this batch for PPO minibatch updates, then collect fresh on-policy data. Export the deterministic actor and its observation transform for evaluation.

Paper exercise: count samples and clip the objective

- 01 Illustrative arithmetic, NOT this run: $16 \text{ environments} \times 128 \text{ steps} = 2,048$ fresh transitions per rollout. Four epochs reuse those transitions; they do not create 8,192 new samples.
- 02 With $\epsilon = 0.2$, ratio = 1.4 and $A = +2$: $\min(2.8, 2.4) = 2.4$. The minimized policy term is -2.4 before averaging and additional loss terms.
- 03 With the same ratio and $A = -2$: $\min(-2.8, -2.4) = -2.8$. Explain why clipping is asymmetric once the sign of the advantage changes.
- 04 Run from repo root: `python3 docs/microduck-training-student-book/learning_lab.py --gae`. This is a deterministic arithmetic lab, not new policy training.

Reward design: what the agent is paid to do · 1/3

- 01 name: track_lin_vel | weight: 2 | sign: reward | definition: exp tracking of body-frame linear velocity to twist_cmd, including vertical velocity error
- 02 name: track_ang_vel | weight: 2 | sign: reward | definition: exp tracking of commanded yaw rate plus roll/pitch angular velocity cost inside the tracking error
- 03 name: upright | weight: 2 | sign: reward | definition: $\exp(-\text{sum}(\text{projected_gravity_xy squared}) / 0.05)$
- 04 name: pose | weight: 1 | sign: reward | definition: $\exp(-\text{sum}(\text{leg joint offsets squared}) / 0.5)$

Reward design: what the agent is paid to do · 2/3

- 05 name: head_pose | weight: 2 | sign: reward | definition: mean per-head-joint $\exp(-(\text{head_error} / 0.5) \text{ squared})$
- 06 name: feet_air_time | weight: 3 | sign: reward | definition: dense payout while a moving foot air-time is between 0.125 s and 0.300 s
- 07 name: action_rate_penalty | weight: lifetime schedule 0.1 at step 0, 0.2 at 12000, 0.4 at 18000, 0.6 at 24000, 0.8 at 30000, and 1.0 at 36000 | sign: penalty | definition: $-\text{weight} * \text{sum}((\text{last_action} - \text{previous_action}) \text{ squared})$
- 08 name: ang_vel_xy_penalty | weight: 0.05 | sign: penalty | definition: $-0.05 * (\text{gyro_x squared} + \text{gyro_y squared})$

Reward design: what the agent is paid to do · 3/3

09

name: bridge_specific_reward | weight: 0 | sign: absent | definition: There is no reward for world x, bridge progress, plank angle, cable tension, destination contact, or bridge_crossed. The bridge checkpoint metadata records reward_weights as an empty override object.

Recorded training recipe · 1/2

- 01 config: transition_budget | value: 32,768 new PPO transitions, 4 environments x 128 steps x 64 updates
- 02 config: ppo_batch | value: 4 environments, 128 steps, 4 minibatches, 2 update epochs
- 03 config: optimizer | value: Adam learning rate 1e-5, maximum gradient norm 0.5, fresh optimizer and fresh seeded critic
- 04 config: discount_and_objective | value: gamma 0.99, GAE lambda 0.95, clip coefficient 0.1, clipped value loss enabled, value coefficient 1.0, normalized advantages

Recorded training recipe · 2/2

- 05 config: exploration | value: initial action standard deviation 0.03, entropy coefficient 0
- 06 config: normalization_and_transfer | value: exact alpha_walking actor and normalizer transfer; parity maximum absolute action error 1.0728836059570312e-06 over 32 samples; observation normalization frozen; reward normalization off
- 07 config: environment | value: XML actuator, dummy backend, bridge curriculum on for training, 20 s maximum episode, seed 7, domain randomization off, observation noise off, action delay off, random yaw off
- 08 config: telemetry_fields | value: phase; env_steps; steps; seconds; optimizer_steps; mean_loss; policy_loss; value_loss; entropy; approximate_kl; clip_fraction; explained_variance; mean_reward; mean_raw_return; returns; lengths

Hands-on commands: inspect, smoke, train · 1/3

01

From .:

```
./scripts/train-bridge.sh train --bridge-curriculum --total-timesteps 32768 --num-envs 4 --  
num-steps 128 --num-minibatches 4 --backend dummy --no-domain-rand --no-obs-noise --no-  
action-delay --no-random-yaw --max-episode-s 20 --seed 7 --output-dir  
rlx/runs/studio/bridge/my-bridge
```

Run the documented bounded training pilot in a fresh output directory.

02

From .:

```
./scripts/train-bridge.sh eval --backend dummy --num-envs 3 --seed 101 --policy  
rlx/runs/studio/bridge/my-bridge/bridge.onnx --eval-steps 1000 --max-episode-s 20 --no-  
domain-rand --no-obs-noise --no-action-delay --no-random-yaw
```

Run the documented unassisted full-horizon crossing evaluation.

Hands-on commands: inspect, smoke, train · 2/3

03

From duck-viewer:

```
npm run dev
```

Open the Studio to inspect the saved failed pilot, evaluation details, and diagnostic render.

04

From .:

```
bash restart-lab.sh --readiness-only
```

Check all scenario routes without claiming that Bridge passed its skill gate.

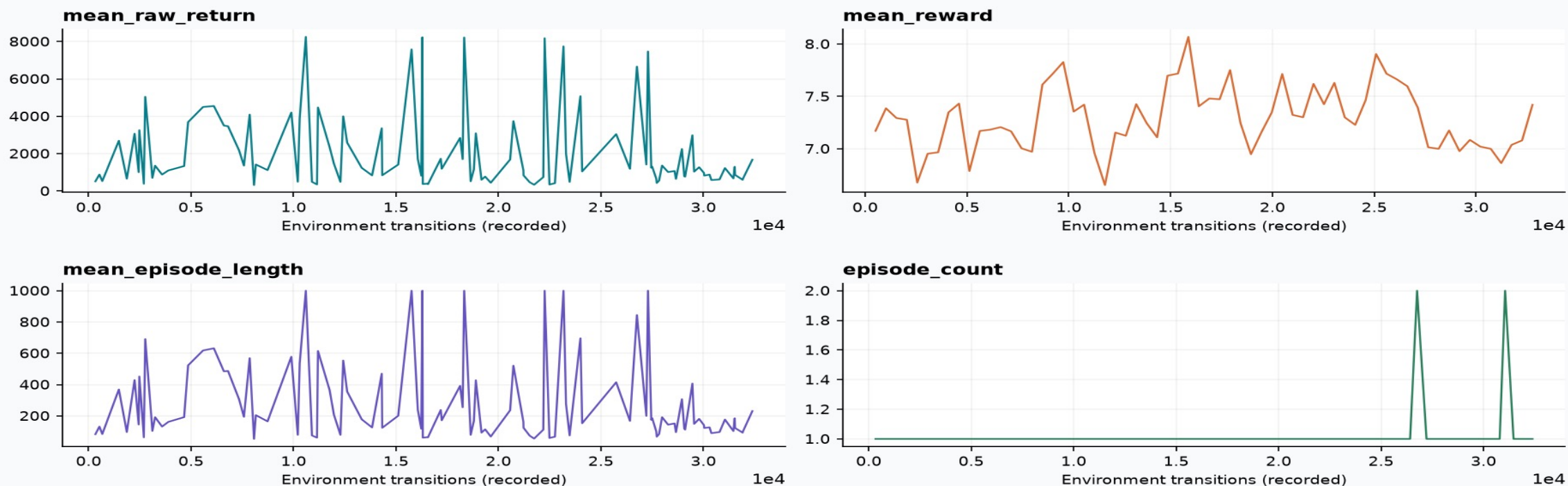
Hands-on commands: inspect, smoke, train · 3/3

05

```
From .:  
node scripts/verify-seven-cases.mjs  
Verify that the custom bridge scene, four tendons, platforms, gantry, and seven-case UI  
integration are present.
```

Reward: does the logged signal match the task?

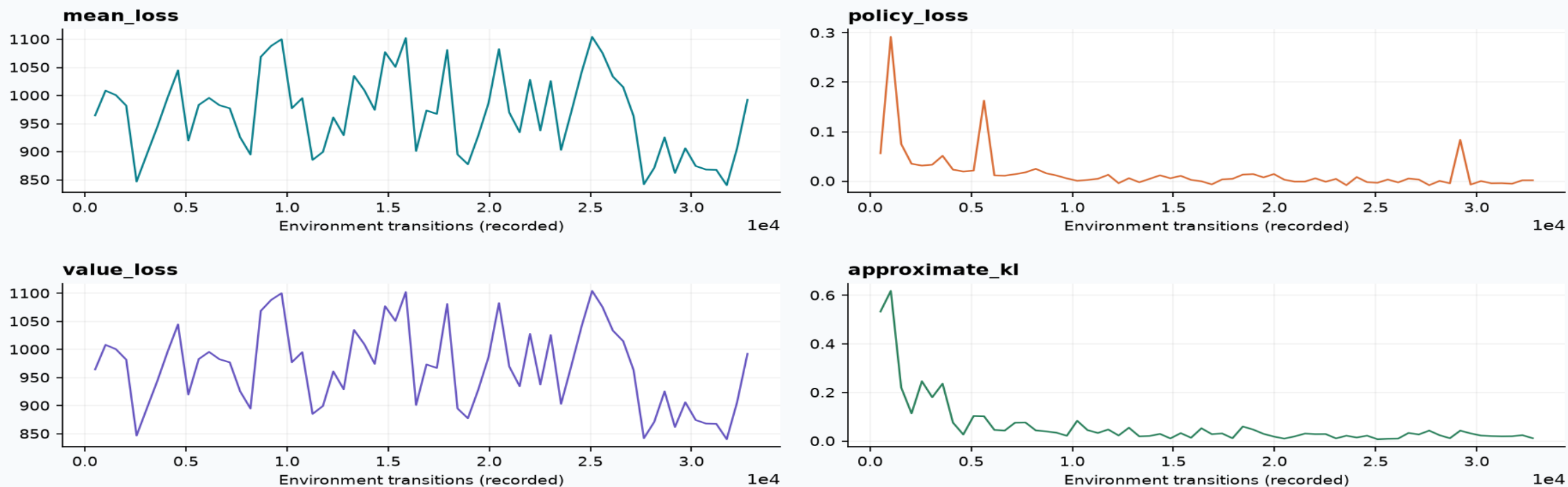
Recorded reward signals — different definitions, not interchangeable



Training reward is an optimization signal, not the acceptance test. See [raw/<case>/](#) for complete recorded fields.

Loss: inspect actor, critic and policy drift

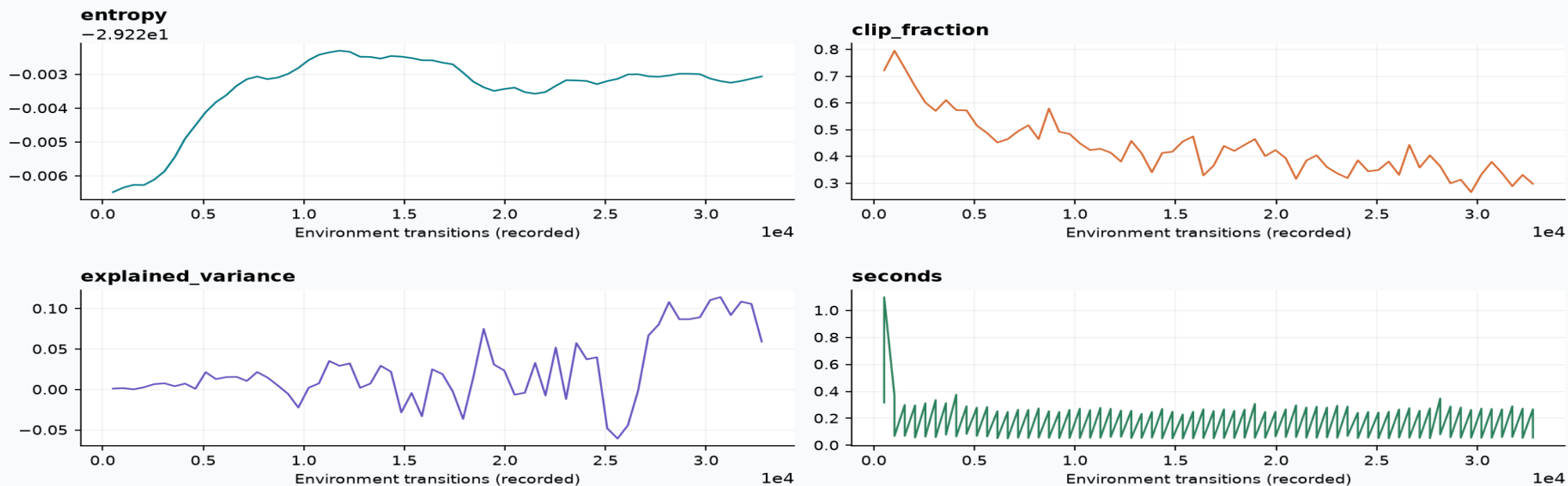
Recorded PPO optimization — raw values, no smoothing



A lower loss does not prove a better robot skill. Loss sign and scale depend on the implementation.

Diagnostics: is the optimizer behaving?

Recorded update diagnostics — absent fields are explicit



Entropy, clipping, explained variance and timing are diagnostics — not a standalone pass/fail certificate.

Metric matrix: numerical audit of this trace · 1/3

- 01 approximate_kl: first 0.5333 → final 0.01181; min 0.007837, max 0.6186; 64 logged samples.
- 02 clip_fraction: first 0.7217 → final 0.2979; min 0.2666, max 0.7949; 64 logged samples.
- 03 entropy: first -29.23 → final -29.22; min -29.23, max -29.22; 64 logged samples.
- 04 explained_variance: first 0.001202 → final 0.05922; min -0.06060, max 0.1143; 64 logged samples.

Metric matrix: numerical audit of this trace · 2/3

05 mean_loss: first 964.7 → final 992.4; min 840.4, max 1104; 64 logged samples.

06 mean_raw_return: first 517.6 → final 1676; min 318.2, max 8254; 110 logged samples.

07 mean_reward: first 7.171 → final 7.420; min 6.653, max 8.069; 64 logged samples.

08 policy_loss: first 0.05633 → final 0.001741; min -0.008248, max 0.2919; 64 logged samples.

Metric matrix: numerical audit of this trace · 3/3

09

value_loss: first 964.7 → final 992.4; min 840.4, max 1104; 64 logged samples.

Verification: define success before looking · 1/3

- 01 Confirm the standard 61-input/14-output actor contract and frozen normalizer parity before training.
- 02 Verify the gantry geoms are non-colliding and cannot create hidden support.
- 03 Verify the curriculum applies forces to the plank only and never changes reward weights.
- 04 Evaluate without --bridge-curriculum and require bridge_assistance=0 for every credited step.

Verification: define success before looking · 2/3

- 05 Require the exact 1,000-step horizon per lane; truncated trailing fragments cannot pass.
- 06 Require at least 1.60 m progress and the contact-validated bridge_crossed flag with both feet on the destination.
- 07 Reject ground contact, non-foot support, missing suspended-plank contact, fall, or incomplete episodes.
- 08 Inspect recipe_metrics fields bridge_crossed, bridge_progress_m, bridge_assistance, bridge_contact_observed, feet_on_bridge, feet_on_end_platform, robot_floor_contact, and nonfoot_support.

Verification: define success before looking · 3/3

09

Watch the full MP4 and count resets; do not combine progress across separate attempts.

10

Separate `pipeline_passed=true` from `skill_status=failed`.

Results: what was actually measured • 1/3

- 01 fact: bridge-studio-02 trained for exactly 32,768 new PPO transitions. | source: `rlx/runs/studio/bridge/bridge-studio-02/result.json`
- 02 fact: The skill evaluation had `pipeline_passed=true` but `passed=false` and `skill_status=failed`. | source: `rlx/runs/studio/bridge/bridge-studio-02/evaluation.json`
- 03 fact: Across 11 observed attempts and trailing fragments, 0 accepted crossings and 0 completed 1,000-step episodes were recorded. | source: `rlx/runs/studio/bridge/bridge-studio-02/evaluation.json`
- 04 fact: Best bridge progress was 1.1749158752636413 m, below the 1.60 m threshold, and `feet_on_end_platform` had maximum 0. | source: `rlx/runs/studio/bridge/bridge-studio-02/evaluation.json`

Results: what was actually measured • 2/3

05 fact: Evaluation assistance remained exactly zero; bridge_assistance min, mean, and max were all 0. | source: `rlx/runs/studio/bridge/bridge-studio-02/evaluation.json`

06 fact: The first PPO update at env_steps 512 recorded mean_loss 964.7162933349609, policy_loss 0.05632817558944225, value_loss 964.6599655151367, approximate_kl 0.5332980752573349, clip_fraction 0.7216796875, and explained_variance 0.0012020504251625264. | source: `rlx/runs/studio/bridge/bridge-studio-02/training-metrics.jsonl`

07 fact: The final PPO update at env_steps 32768 recorded mean_loss 992.3935241699219, policy_loss 0.0017409028951078653, value_loss 992.3917770385742, approximate_kl 0.011814007302746177, clip_fraction 0.2978515625, and explained_variance 0.05922189893301022. | source: `rlx/runs/studio/bridge/bridge-studio-02/training-metrics.jsonl`

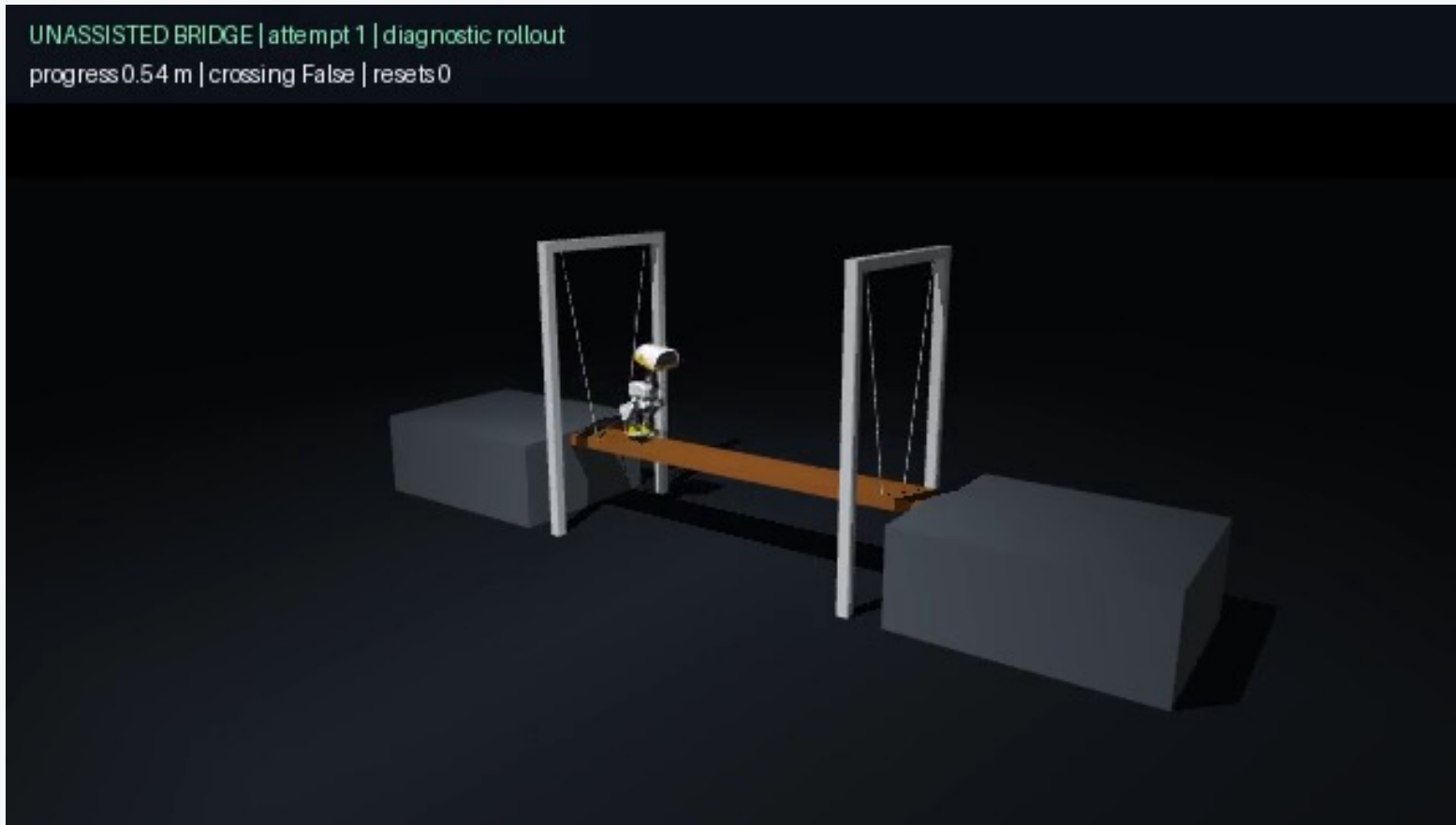
08 fact: The retained 20-second diagnostic video contains two explicit resets and later partial plank progress; it is not one continuous crossing. | source: `docs/bridge-showcase/RESULTS.md`

Results: what was actually measured • 3/3

09

fact: The render MP4 is hash-bound as SHA256
7386c84eb8065d2f4ac2b297ba5b082dc658648657e4190a9491872bab6e50a6. |
source: rlx/runs/studio/bridge/bridge-studio-02/render/evidence.json

Watch the actual saved rollout



BEFORE PLAY

Predict one success signal and one failure signal.

AFTER PLAY

Which claims does this clip support? What remains untested?

20.0 s saved MP4 • Click in PowerPoint Slide Show • Backup: videos/bridge-rollout.mp4

Limits and failure analysis • 1/2

- 01 The bridge crossing failed. Partial progress and a valid scene do not establish the requested skill.
- 02 The navigation commander uses privileged world position and heading, so the complete protocol includes more than the learned actor.
- 03 The actor is blind to plank state and must adapt only through IMU, joints, previous action, and generated command slots.
- 04 The pilot is an actor warm-start from alpha_walking with a fresh critic and optimizer, not exact continuation of upstream training.

Limits and failure analysis • 2/2

- 05 Curriculum completion was not established; 32,768 transitions do not imply graduation to the final unassisted rung.
- 06 The retained evaluation disabled randomization, noise, delay, and random yaw and used only three vector lanes.
- 07 Plank, friction, tendon, and cable values are plausible simulator parameters, not measurements or a certified construction plan.
- 08 No hardware-safe crossing or sim-to-real result is claimed.

Student lab: change one variable, keep the gates · 1/3

- 01 Draw the bridge side view to scale from the platform, plank, and anchor coordinates.
- 02 Compute the nominal cable length from the 0.113 m lateral and 0.545 m vertical offsets.
- 03 Mark every privileged commander input and every value actually delivered to the 61-observation actor.
- 04 Explain why `bridge_progress_m` is valid for evaluation but intentionally absent from the reward.

Student lab: change one variable, keep the gates · 2/3

- 05 Calculate batch size, number of updates, and optimizer minibatch size for the 32,768-transition run.
- 06 Plot the first and final update loss fields and explain why neither proves a crossing.
- 07 Read the 11 evaluation attempts and classify each failure as fall, incomplete horizon, missing plank contact, floor contact, or insufficient progress.
- 08 Watch ep0.mp4, timestamp both resets, and identify the farthest continuous attempt.

Student lab: change one variable, keep the gates · 3/3

09

Design one physics/spawn curriculum change that preserves the reward and final evaluation gate.

10

Write a result slide whose headline states failure clearly while preserving the useful environment and exploration evidence.

Submit an experiment, not just a video

- 01** Before training: write one hypothesis, the changed parameter, the fixed baseline, seed list, acceptance criteria and time/transition budget.
- 02** During training: save the exact command, environment versions, initialization, checkpoint and raw reward/loss/KL metrics. Never overwrite the saved reference run.
- 03** After training: evaluate the frozen exported policy on matched seeds; include all failures, aggregate metrics, the full rollout video and a source-bound result table.
- 04** Rubric / 10: reproducibility 2; environment and reward explanation 2; PPO and metrics interpretation 2; honest verification 2; one evidence-based next experiment 2.

Source map and further study · 1/2

01

microduck_local/AGENTS.md

02

docs/student-book/microduck-lab-student-book.md

03

docs/basketball-bridge-book/chapters/en/03-ppo.md

04

docs/basketball-bridge-book/chapters/en/05-bridge.md

Source map and further study · 2/2

05

`docs/microduck-training-student-book/chapters/02a-ppo-code-lab.md`

06

`docs/microduck-training-student-book/chapters/02b-gae-workshop.md`

07

`scripts/scenarios/README.md`

08

`README.md`